

What is Rekall?

Rekall is like a **digital detective's tool** that helps investigators look deep into a computer's memory (RAM) to find out what was going on behind the scenes—even if a hacker tried to hide their tracks.

What Can Rekall Do?

- **Spot hidden or suspicious programs** running in memory—even if they don't show up normally.
- **Find malware** that never touches the hard drive (so-called **fileless malware**).
- **See what commands were typed** into the command prompt or PowerShell.
- **Check for stolen passwords** or login attempts.
- **Detect if someone tried to stay hidden** by injecting their code into legit processes like

Why Use Rekall?

- It's **fast** and powerful.
- It works on **Windows, Linux, and Mac** memory dumps.
- It can be used for **cybercrime investigations, malware analysis**, or just to understand what was happening on a machine when things went wrong.

How to install REKALL? And How to Run Rekall?

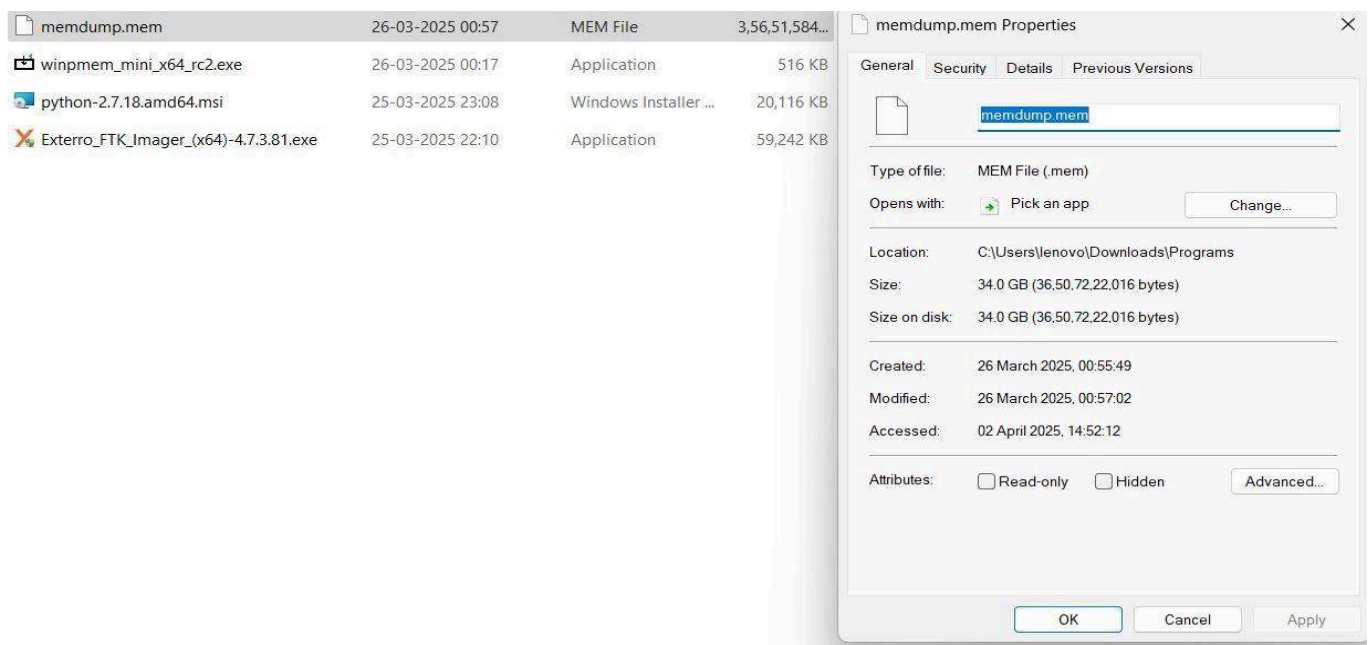
Step 1: Download REKALL FROM - <https://github.com/google/rekall>

```
Microsoft Windows [Version 10.0.22631.5126]
(c) Microsoft Corporation. All rights reserved.

C:\Users\lenovo> cd C:\Program Files\Rekall

C:\Program Files\Rekall>
```

Step 2: Run the REKALL file in CMD (command prompt).



Step 3: Create Forensic Ram Image by using 'winpmem'.

```
C:\Users\lenovo> cd C:\Program Files\Rekall  
C:\Program Files\Rekall> rekall -f
```

```
C:\Users\lenovo> cd C:\Program Files\Rekall  
C:\Program Files\Rekall> rekall -f "C:\Users\lenovo\Downloads\Programs\memdump.mem" pstree
```

Step 4: open CMD, run rekall than use rekall command 'rekall -f'.

```
C:\Users\lenovo> cd C:\Program Files\Rekall  
C:\Program Files\Rekall> rekall -f "C:\Users\lenovo\Downloads\Programs\memdump.mem"
```

Step 5: Copy the file by using 'copy as path' paste in CMD.

Step 6: Write recall analysing commands.

```

C:\Users\lenovo> cd C:\Program Files\Rekall

C:\Program Files\Rekall> rekall -f "C:\Users\lenovo\Downloads\Programs\memdump.mem" pstree
2025-04-04 16:18:25,215:WARNING:rekall.1:Inventory for repository "http://profiles.rekall-forensic.com" seems malformed. Are you behind a captive portal or proxy? If this is a custom repository, did you forget to create an inventory? You must use the tools/profiles/build_profile_repo.py tool with the --inventor y flag.
2025-04-04 16:18:25,218:WARNING:rekall.1:Repository http://profiles.rekall-forensic.com will be disabled.

```

_EPROCESS	ppid	thd_count	hnd_count	create_time
0x8d8a476ff040 System (4)	0	445	-	2025-03-25 19:17:15Z
0x8d8a478d6040 Secure System (204)	4	0	-	2025-03-25 19:17:08Z
0x8d8a478e8040 Registry (236)	4	4	-	2025-03-25 19:17:08Z
0x8d8a54277040 smss.exe (880)	4	2	-	2025-03-25 19:17:15Z
0x8d8a57f13080 MemCompression (3604)	4	66	-	2025-03-25 19:17:27Z
0x8d8a61d7e100 csrss.exe (1240)	1000	17	-	2025-03-25 19:17:24Z
0x8d8a647ee100 wininit.exe (1356)	1000	2	-	2025-03-25 19:17:26Z
0x8d8a577790c0 services.exe (1428)	1356	9	-	2025-03-25 19:17:26Z
0x8d8a5e78a0c0 AISControlServ (900)	1428	2	-	2025-03-25 19:17:29Z
0x8d8a5e79b0c0 mDNSResponder. (1348)	1428	2	-	2025-03-25 19:17:29Z
0x8d8a64922140 csrss.exe (1376)	1348	16	-	2025-03-25 19:17:26Z
0x8d8a649be100 winlogon.exe (1500)	1348	5	-	2025-03-25 19:17:26Z
0x8d8a578240c0 fontdrvhost.ex (1696)	1500	5	-	2025-03-25 19:17:26Z
0x8d8a579ce0c0 dwm.exe (2432)	1500	34	-	2025-03-25 19:17:26Z
0x8d8a5daac0c0 userinit.exe (6632)	1500	0	-	2025-03-25 19:17:28Z
0x8d8a5dbbd0c0 explorer.exe (6708)	6632	196	-	2025-03-25 19:17:28Z
0x8d8a6d942080 OneDrive.exe (3248)	6708	46	-	2025-03-25 19:17:42Z
0x8d8a6e5f6080 IDMan.exe (12296)	6708	5	-	2025-03-25 19:17:47Z
0x8d8a6d9980c0 vgtray.exe (20916)	6708	4	-	2025-03-25 19:17:42Z
0x8d8a6bc66080 browser_assist (21144)	6708	8	-	2025-03-25 19:17:44Z
0x8d8a6b8e4080 browser_assist (9004)	21144	7	-	2025-03-25 19:17:44Z
0x8d8a6d9020c0 SecurityHealth (21148)	6708	3	-	2025-03-25 19:17:40Z
0x8d8a6bf8e0c0 uTorrent.exe (21192)	6708	40	-	2025-03-25 19:17:45Z
0x8d8a6b2f3080 utorrentie.exe (21556)	21192	7	-	2025-03-25 19:17:45Z
0x8d8a6b6b9080 utorrentie.exe (21612)	21192	7	-	2025-03-25 19:17:46Z
0x8d8a6dc220c0 msedgewebview2 (22216)	21612	54	-	2025-03-25 19:17:46Z
0x8d8a6dc56080 msedgewebview2 (22544)	22216	7	-	2025-03-25 19:17:46Z
0x8d8a6e0df080 msedgewebview2 (22832)	22216	51	-	2025-03-25 19:17:46Z
0x8d8a6e0e0080 msedgewebview2 (22840)	22216	18	-	2025-03-25 19:17:46Z
0x8d8a6e0da080 msedgewebview2 (22880)	22216	7	-	2025-03-25 19:17:46Z
0x8d8a6e291080 msedgewebview2 (23012)	22216	14	-	2025-03-25 19:17:46Z
0x8d8a6e293100 msedgewebview2 (23072)	22216	23	-	2025-03-25 19:17:46Z

Step 7. Then press 'ENTER'.

How to get memory dump file ?

Step 1 – download by using github memory (many file are there)

Step 2 – firstly download 'winpmem'

Step 3 – use winpmem in cmd write the location and run it.

Step 4 – Wright 'pip install the winpmem'

Step 5 – than install the python 2 to use the winpmem to create the memory dump file

Step 6 – create the memory dump file and download it

Step 7 – first Wright the location to save the file in your device

Step 8 – name the file

Step 9 – then press enter to download the memory dump file

Step 10 – the file take many time to install (depends on ram)

Step 11 – this file are .mem file , .raw file.

Step 12 – choose to protect your file with password is your choice

Many error face to download the memory dump file (.raw , .mem) –

1. pip installation is failing all the time, tried with older pip versions, latest version, tried with python 2 and python 3, but installs unsuccessful.
2. Its not supporting the latest python version.
3. This will give the error in installing the file because of the ram were 32 GB so this will took much time to install.
4. Many time were failed to install.
5. Total of time to install the file is 3 hour max to max.
6. Its showing warnings.
7. One time the file were stuck.
8. Two to three time the file were showing the protection or warning to install
9. In last the file were installed completely but not working properly.
10. Commands were not work in file.
11. 1 more hour later than the file were proper work the command work properly.

```

Using cached rekall_efilter-1.6.0-py3-none-any.whl.metadata (639 bytes)
WARNING: Ignoring version 1.6.0 of rekall-efilter since it has invalid metadata:
Requested rekall-efilter<1.7,>=1.6 from https://files.pythonhosted.org/packages/42/ce/43e12f2bd6babb9d130e960ea8177cd7dc
f87f54ae625a7a70b081eca3d1/rekall_efilter-1.6.0-py3-none-any.whl (from rekall-core<1.8,>=1.7.0rc1->rekall==1.7.2rc1) has
invalid metadata: Expected matching RIGHT_PARENTHESIS for LEFT_PARENTHESIS, after version specifier
pytz (>=2011k)

Please use pip<24.1 if you need to use this version.
ERROR: Cannot install rekall because these package versions have conflicting dependencies.

The conflict is caused by:
  rekall-core 1.7.2rc1 depends on rekall-efilter<1.7 and >=1.6
  rekall-core 1.7.1 depends on rekall-efilter<1.7 and >=1.6

To fix this you could try to:
1. loosen the range of package versions you've specified
2. remove package versions to allow pip to attempt to solve the dependency conflict

ERROR: ResolutionImpossible: for help visit https://pip.pypa.io/en/latest/topics/dependency-resolution/#dealing-with-dep
endency-conflicts

C:\Users\lenovo\Downloads\Compressed\rekall-master>C:\Users\lenovo\Downloads\Compressed\rekall-masterC:\Users\lenovo\Down
loads\Compressed\rekall-master

```

```

Collecting pypiwin32==219
  Downloading pypiwin32-219.zip (4.8 MB)
    4.8/4.8 MB 942.4 kB/s eta 0:00:00
  Preparing metadata (setup.py) ... error
  error: subprocess-exited-with-error

  × python setup.py egg_info did not run successfully.
  exit code: 1
  [16 lines of output]
  Traceback (most recent call last):
    File "<string>", line 2, in <module>
      exec(compile('
  # This is <pip-setuptools-caller> -- a caller that pip uses to run setup.py
  ...<31 lines>...
  exec(compile(setup_py_code, filename, "exec"))
  ''' % ('C:\\Users\\lenovo\\AppData\\Local\\Temp\\pip-install-9yzxitxv\\pypiwin32_03d78755122c49209b2a39c6e0838
  0c4\\setup.py',), "<pip-setuptools-caller>", "exec"))
  A new release of pip is available: 23.3.1 -> 25.0.1
  To update, run: python.exe -m pip install --upgrade pip

```

How It Works in Real Life?

```

Preparing metadata (setup.py) ... done
ERROR: Cannot install rekall because these package versions have conflicting dependencies.

The conflict is caused by:
  rekall-core 1.7.2rc1 depends on pypiwin32==220; sys_platform == "win32"
  rekall-core 1.7.1 depends on pypiwin32==220; sys_platform == "win32"

To fix this you could try to:
1. loosen the range of package versions you've specified
2. remove package versions to allow pip attempt to solve the dependency conflict

ERROR: ResolutionImpossible: for help visit https://pip.pypa.io/en/latest/topics/dependency-resolution/#dealing-with-dep
endency-conflicts

[notice] A new release of pip is available: 23.3.1 -> 25.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\lenovo\Downloads\Compressed\rekall-master>

```

A

forensic analyst is investigating a potential security breach on a Windows machine at SkyTech Financial, a leading investment firm. The company's security operations center (SOC) detected unusual network traffic originating from an executive's laptop at 2:00 AM—outside business hours. No malware was found on disk, raising concerns of a fileless attack.

To uncover evidence, the forensic analyst uses Rekall, a powerful memory forensics framework, to examine a RAM dump for traces of malware, unauthorized processes, and unusual behavior.

There are a lot of things we can investigate to solve the scenario described earlier but for lab purposes we will be finding answers to the following 10 questions.

1. What processes were running at the time of the security breach?

Command:

rekall -f memory_memdumb.mem pslist

```
C:\Program Files\Rekall> rekall -f "C:\Users\lenovo\Downloads\Programs\memdump.mem" pslist
2025-04-02 11:00:41,278:WARNING:rekall.1:Inventory for repository "http://profiles.rekall-forensic.com" seems malformed. Are you behind a captive portal or proxy? If this is a custom repository, did you forget to create an inventory? You must use the tools/profiles/build_profile_repo.py tool with the --inventory flag.
2025-04-02 11:00:41,278:WARNING:rekall.1:Repository http://profiles.rekall-forensic.com will be disabled.
```

_EPROCESS	name	pid	ppid	thread_count	handle_count	session_id	wow64	process_create_time	process_exit_time
0x8d8a476ff040	System	4	0	445	-	-	False	2025-03-25 19:17:15Z	-
0x8d8a478d6040	Secure System	204	4	0	-	-	False	2025-03-25 19:17:08Z	-
0x8d8a478e8040	Registry	236	4	4	-	-	False	2025-03-25 19:17:08Z	-
0x8d8a54277040	smss.exe	880	4	2	-	-	False	2025-03-25 19:17:15Z	-
0x8d8a5e78a0c0	AISControlServ	900	1428	2	-	0	False	2025-03-25 19:17:29Z	-
0x8d8a716db0c0	LenovoVantage-	1228	7220	0	-	1	False	2025-03-25 19:22:49Z	2025-03-25 19:22:57Z
0x8d8a61d7e100	csrss.exe	1240	1000	17	-	0	False	2025-03-25 19:17:24Z	-
0x8d8a5e79b0c0	mDNSResponder.	1348	1428	2	-	0	False	2025-03-25 19:17:29Z	-
0x8d8a47ee100	wininit.exe	1356	1000	2	-	0	False	2025-03-25 19:17:26Z	-
0x8d8a64922140	csrss.exe	1376	1348	16	-	1	False	2025-03-25 19:17:26Z	-
0x8d8a577790c0	services.exe	1428	1356	9	-	0	False	2025-03-25 19:17:26Z	-
0x8d8a5778a0c0	LsaIso.exe	1436	1356	1	-	0	False	2025-03-25 19:17:26Z	-
0x8d8a5779b0c0	lsass.exe	1492	1356	12	-	0	False	2025-03-25 19:17:26Z	-
0x8d8a649be100	winlogon.exe	1500	1348	5	-	1	False	2025-03-25 19:17:26Z	-
0x8d8a578df0c0	svchost.exe	1592	1428	4	-	0	False	2025-03-25 19:17:26Z	-
0x8d8a578130c0	svchost.exe	1660	1428	21	-	0	False	2025-03-25 19:17:26Z	-
0x8d8a710ac080	Locator.exe	1668	1428	2	-	0	False	2025-03-25 19:22:37Z	-
0x8d8a578350c0	fontdrvhost.ex	1688	1356	5	-	0	False	2025-03-25 19:17:26Z	-
0x8d8a578240c0	fontdrvhost.ex	1696	1500	5	-	1	False	2025-03-25 19:17:26Z	-
0x8d8a52d07080	ms-teams.exe	1740	23916	103	-	1	False	2025-03-25 19:18:01Z	-
0x8d8a578460c0	WUDFHost.exe	1792	1428	5	-	0	False	2025-03-25 19:17:26Z	-
0x8d8a578570c0	svchost.exe	1812	1428	11	-	0	False	2025-03-25 19:17:26Z	-
0x8d8a70295080	backgroundTask	1868	1660	4	-	1	False	2025-03-25 19:22:37Z	-
0x8d8a578680c0	svchost.exe	1876	1428	6	-	0	False	2025-03-25 19:17:26Z	-
0x8d8a6f482080	SEGameTool.exe	1932	17956	5	-	1	False	2025-03-25 19:22:41Z	-
0x8d8a5788a0c0	svchost.exe	1980	1428	2	-	0	False	2025-03-25 19:17:26Z	-
0x8d8a578ac0c0	svchost.exe	2016	1428	4	-	0	False	2025-03-25 19:17:26Z	-
0x8d8a578bd0c0	svchost.exe	2024	1428	8	-	0	False	2025-03-25 19:17:26Z	-

Analysis:

Suspicious third-party binary (SEGameTool.exe) that may be a loader, injector, or tool used in post-exploitation.

2. Are there any injected or hollowed processes?

Command:

```
rekall -f memory_memdumb.mem malfind
```

```
Process: msedgewebview2 Pid: 24384 Address: 0x7ff918160000
Vad Tag: VadS Protection: EXECUTE_READWRITE
PrivateMemory: 1, Protection: 6

0x7ff918160000 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ..... vad_0x7ff918160000
0x7ff918160010 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0x7ff918160020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0x7ff918160030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....

----- vad_0x7ff918160000 -----
0x7ff918160000 0x0 0000 add byte ptr [rax], al
0x7ff918160002 0x2 0000 add byte ptr [rax], al
0x7ff918160004 0x4 0000 add byte ptr [rax], al
0x7ff918160006 0x6 0000 add byte ptr [rax], al
0x7ff918160008 0x8 0000 add byte ptr [rax], al
0x7ff91816000a 0xa 0000 add byte ptr [rax], al
0x7ff91816000c 0xc 0000 add byte ptr [rax], al
0x7ff91816000e 0xe 0000 add byte ptr [rax], al
0x7ff918160010 0x10 0000 add byte ptr [rax], al
0x7ff918160012 0x12 0000 add byte ptr [rax], al
0x7ff918160014 0x14 0000 add byte ptr [rax], al
0x7ff918160016 0x16 0000 add byte ptr [rax], al
0x7ff918160018 0x18 0000 add byte ptr [rax], al
0x7ff91816001a 0x1a 0000 add byte ptr [rax], al
0x7ff91816001c 0x1c 0000 add byte ptr [rax], al
0x7ff91816001e 0x1e 0000 add byte ptr [rax], al
0x7ff918160020 0x20 0000 add byte ptr [rax], al
0x7ff918160022 0x22 0000 add byte ptr [rax], al
0x7ff918160024 0x24 0000 add byte ptr [rax], al
0x7ff918160026 0x26 0000 add byte ptr [rax], al
0x7ff918160028 0x28 0000 add byte ptr [rax], al
0x7ff91816002a 0x2a 0000 add byte ptr [rax], al
0x7ff91816002c 0x2c 0000 add byte ptr [rax], al
0x7ff91816002e 0x2e 0000 add byte ptr [rax], al
0x7ff918160030 0x30 0000 add byte ptr [rax], al
0x7ff918160032 0x32 0000 add byte ptr [rax], al
0x7ff918160034 0x34 0000 add byte ptr [rax], al
0x7ff918160036 0x36 0000 add byte ptr [rax], al
0x7ff918160038 0x38 0000 add byte ptr [rax], al
0x7ff91816003a 0x3a 0000 add byte ptr [rax], al
0x7ff91816003c 0x3c 0000 add byte ptr [rax], al
0x7ff91816003e 0x3e 0000 add byte ptr [rax], al
0x7ff918160040 0x40 0000 add byte ptr [rax], al
```

Analysis: The msedgewebview2 process has a suspicious memory region that:

- Is marked EXECUTE_READWRITE (bad sign).
- Contains meaningless or empty instructions (add byte ptr [rax], al)
— commonly seen in hollowing.

- Has private memory allocations that aren't backed by a file.

This highlights suspicion of presence of hollowed memory

3. What DLLs were loaded by suspicious processes?

Command:

rekall -f memory_memdumb.mem dlllist

```
Command line : "C:\Program Files\WindowsApps\MicrosoftWindows.Client.WebExperience_525.1301.30.0_x64__cw5n1h2txyewy\Dashboard\Widgets.exe" -ServerName:Micro
soft.Windows.DashboardServer
```

```
-----
0x7ff7e8f50000 0x2b9000 LoadReasonDynamicLoad C:\Program Files\WindowsApps\MicrosoftWindows.Client.WebExperience_525.1301.30.0_x64__cw5n1h2
txyewy\Dashboard\Widgets.exe
0x7ff919eb0000 0x217000 LoadReasonStaticDependency C:\Windows\SYSTEM32\ntdll.dll
0x7ff918170000 0xc4000 LoadReasonDynamicLoad C:\Windows\System32\KERNEL32.DLL
0x7ff917680000 0x3d1000 LoadReasonStaticDependency C:\Windows\System32\KERNELBASE.dll
0x7ff919c00000 0x1b1000 LoadReasonStaticDependency C:\Windows\System32\USER32.dll
0x7ff916ff0000 0x26000 LoadReasonStaticDependency C:\Windows\System32\win32u.dll
0x7ff919500000 0x29000 LoadReasonStaticDependency C:\Windows\System32\GDI32.dll
0x7ff917560000 0x11b000 LoadReasonStaticDependency C:\Windows\System32\gdi32full.dll
0x7ff9170d0000 0x9a000 LoadReasonStaticDependency C:\Windows\System32\msvcp_win.dll
0x7ff917a60000 0x111000 LoadReasonStaticDependency C:\Windows\System32\ucrtbase.dll
0x7ff918240000 0x390000 LoadReasonStaticDependency C:\Windows\System32\combase.dll
0x7ff919270000 0x114000 LoadReasonStaticDependency C:\Windows\System32\RPCRT4.dll
0x7ff919af0000 0x10a000 LoadReasonStaticDependency C:\Windows\System32\shcore.dll
0x7ff8beab0000 0xc000 LoadReasonStaticDependency C:\Program Files\WindowsApps\Microsoft.VCLibs.140.00_14.0.33519.0_x64__8wekyb3d8bbwe\VCRUNTIM
E140_1_APP.dll
0x7ff8b22d0000 0x1d000 LoadReasonStaticDependency C:\Program Files\WindowsApps\Microsoft.VCLibs.140.00_14.0.33519.0_x64__8wekyb3d8bbwe\VCRUNTIM
E140_APP.dll
0x7ff8b0cd0000 0x4a000 LoadReasonStaticDependency C:\Program Files\WindowsApps\Microsoft.VCLibs.140.00_14.0.33519.0_x64__8wekyb3d8bbwe\CONCRT14
0_APP.dll
0x7ff8b0d20000 0x8c000 LoadReasonStaticDependency C:\Program Files\WindowsApps\Microsoft.VCLibs.140.00_14.0.33519.0_x64__8wekyb3d8bbwe\MSVCP140
_APP.dll
0x7ff8b0ca0000 0x2d000 LoadReasonStaticDependency C:\Program Files\WindowsApps\MicrosoftWindows.Client.WebExperience_525.1301.30.0_x64__cw5n1h2
txyewy\Dashboard\Webview2Loader.dll
0x7ff918730000 0x31000 LoadReasonDynamicLoad C:\Windows\System32\IMM32.DLL
0x7ff916000000 0x18000 LoadReasonDelayloadDependency C:\Windows\SYSTEM32\kernel.appcore.dll
0x7ff9185d0000 0xa7000 LoadReasonStaticDependency C:\Windows\System32\msvcrt.dll
0x7ff9174e0000 0x7b000 LoadReasonDelayloadDependency C:\Windows\System32\bcryptPrimitives.dll
0x7ff918680000 0xb0000 LoadReasonDelayloadDependency C:\Windows\System32\clbcatq.dll
0x7ff8f1220000 0xc6000 LoadReasonDynamicLoad C:\Windows\System32\Windows.Web.dll
0x7ff919440000 0xb1000 LoadReasonDynamicLoad C:\Windows\System32\ADVAPI32.dll
0x7ff919390000 0xa7000 LoadReasonStaticDependency C:\Windows\System32\sechost.dll
0x7ff917020000 0x28000 LoadReasonStaticDependency C:\Windows\System32\bcrypt.dll
0x7ff917b80000 0x1a1000 LoadReasonDynamicLoad C:\Windows\System32\ole32.dll
0x7ff918970000 0x888000 LoadReasonDynamicLoad C:\Windows\System32\SHELL32.dll
0x7ff9173a0000 0x13f000 LoadReasonStaticDependency C:\Windows\System32\wintypes.dll
0x7ff90c0a0000 0x1a000 LoadReasonDelayloadDependency C:\Windows\SYSTEM32\windows.staterepositorycore.dll
0x7ff8b2350000 0x10c000 LoadReasonDynamicLoad C:\Windows\System32\ShellCommonCommonProxyStub.dll
0x7ff914620000 0xb3000 LoadReasonDynamicLoad C:\Windows\system32\uxtheme.dll
```

Analysis:

Highlighted Suspicious Entry:

DashboardWebView2Loader.dll

Behavior: Likely DLL side-loading or disguised payload loader.

4. Are there any signs of persistence mechanisms in memory?

```
C:\Program Files\Rekal> rekal -f "C:\Users\lenovo\Downloads\Programs\memdump.mem" autoruns
usage: rekal [-p PROFILE] [-v] [-q] [--debug] [--output_style {concise,full}]
             [--logging_level {DEBUG,INFO,WARNING,ERROR,CRITICAL}]
             [--log_domain [{PageTranslation} [{PageTranslation} ...]]]
             [--pager PAGER] [--paging_limit PAGING_LIMIT]
             [--colors {auto,yes,no}] [-F FORMAT]
             [--plugin [PLUGIN [PLUGIN ...]]] [-h] [--cache CACHE]
             [--repository_path [REPOSITORY_PATH [REPOSITORY_PATH ...]]]
             [-f FILENAME] [--buffer_size BUFFER_SIZE] [--output OUTPUT]
             [--max_collector_cost MAX_COLLECTOR_COST] [--home HOME]
             [--logging_format LOGGING_FORMAT]
             [--performance {normal,fast,thorough}] [--live LIVE]
             [--cpu_quota CPU_QUOTA] [--load_quota LOAD_QUOTA] [--dtb DTB]
             [-o FILE_OFFSET] [--ept EPT [EPT ...]] [--timezone TIMEZONE]
             [--cache_dir CACHE_DIR]
             [--highlighting_style {default,emacs,friendly,colorful,autumn,murphy,manni,monokai,perldoc,pastie,borland,trac,native,fruity,bw,vim,vs,tango,rr
t,xcode,igor,paraiso-light,paraiso-dark,lovelace,algol,algol_nu,arduino,rainbow_dash,abap}]
             [--name_resolution_strategies [{Module,Symbol,Export} [{Module,Symbol,Export} ...]]]
             [--autodetect_build_local_tracked [AUTODETECT_BUILD_LOCAL_TRACKED [AUTODETECT_BUILD_LOCAL_TRACKED ...]]]
             [--pagefile [PAGEFILE [PAGEFILE ...]]]
             [--autodetect {nt_index,pe,rsds,windows_kernel_file,linux_index,linux,osx,ntfs,tsk} [{nt_index,pe,rsds,windows_kernel_file,linux_index,linux,os
x,ntfs,tsk} ...]]
             [--autodetect_threshold AUTODETECT_THRESHOLD]
             [--autodetect_build_local {full,basic,none}]
             [--autodetect_scan_length AUTODETECT_SCAN_LENGTH] [--version]
             [--agent_configuration AGENT_CONFIGURATION] [-]
rekal: error: unrecognized arguments: autoruns
```

Command:

rekall -f memory_memdumb.mem autoruns

Analysis:

no signs of persistence mechanisms could be confirmed

5. Were any suspicious kernel modules loaded?

Command:

rekall -f memory_memdumb.mem modules

```
C:\Program Files\Rekal> rekall -f "C:\Users\lenovo\Downloads\Programs\memdump.mem" modules
2025-04-02 15:30:01,203:WARNING:rekall.1:Inventory for repository "http://profiles.rekall-forensic.com" seems malformed. Are you behind a captive portal or
proxy? If this is a custom repository, did you forget to create an inventory? You must use the tools/profiles/build_profile_repo.py tool with the --inventor
y flag.
2025-04-02 15:30:01,223:WARNING:rekall.1:Repository http://profiles.rekall-forensic.com will be disabled.
_LDR_DATA_TABLE_ENTRY      name      base      size      path
-----
0x8d8a47658510      ntoskrnl.exe      0xf80158200000      0x1047000 C:\Windows\system32\ntoskrnl.exe
0x8d8a47658960      hal.dll      0xf80155790000      0x6000 C:\Windows\system32\hal.dll
0x8d8a47663c20      kdcom.dll      0xf801557a0000      0xb000 C:\Windows\system32\kd.dll
0x8d8a47698270      CLFS.SYS      0xf801557e0000      0x6f000 C:\Windows\System32\drivers\CLFS.SYS
0x8d8a47698400      tm.sys      0xf801557b0000      0x29000 C:\Windows\System32\drivers\tm.sys
0x8d8a47698610      PSHED.dll      0xf80155850000      0x1b000 C:\Windows\system32\PSHED.dll
0x8d8a47699010      BOOTVID.dll      0xf80155870000      0xd000 C:\Windows\system32\BOOTVID.dll
0x8d8a476991d0      FLTMRG.SYS      0xf801559a0000      0x7a000 C:\Windows\System32\drivers\FLTMRG.SYS
0x8d8a476993b0      msrpc.sys      0xf80155a50000      0x62000 C:\Windows\System32\drivers\msrpc.sys
0x8d8a47699590      ksecdd.sys      0xf80155a20000      0x2c000 C:\Windows\System32\drivers\ksecdd.sys
0x8d8a47699760      clipsp.sys      0xf80155880000      0x115000 C:\Windows\System32\drivers\clipsp.sys
0x8d8a47699980      cmimext.sys      0xf80155ac0000      0x12000 C:\Windows\System32\drivers\cmimext.sys
0x8d8a47699b50      werkernel.sys      0xf80155ae0000      0x16000 C:\Windows\System32\drivers\werkernel.sys
0x8d8a47699d30      ntosext.sys      0xf80155b00000      0xc000 C:\Windows\System32\drivers\ntosext.sys
0x8d8a4769a010      CI.dll      0xf8015b400000      0xf7000 C:\Windows\system32\CI.dll
0x8d8a4769a200      cng.sys      0xf8015b500000      0xbe000 C:\Windows\System32\drivers\cng.sys
0x8d8a4769a3d0      Wdf01000.sys      0xf8015b5c0000      0xc7000 C:\Windows\system32\drivers\Wdf01000.sys
0x8d8a4769a5c0      WppRecorder.sys      0xf80155b30000      0x13000 C:\Windows\system32\drivers\WppRecorder.sys
0x8d8a4769a7a0      WDFLDR.SYS      0xf80155b10000      0x17000 C:\Windows\system32\drivers\WDFLDR.SYS
0x8d8a4769a970      PRM.sys      0xf80155b50000      0xe000 C:\Windows\System32\DriverStore\FileRepository\prm.inf_amd64_de435dc5c75d64a5\PRM.s
ys
0x8d8a4769ab40      acpiex.sys      0xf80155f70000      0x27000 C:\Windows\System32\Drivers\acpiex.sys
0x8d8a4769ad10      ACPI.sys      0xf8015b690000      0xb8000 C:\Windows\System32\drivers\ACPI.sys
0x8d8a476a8050      WMILIB.SYS      0xf80155fa0000      0xc000 C:\Windows\System32\drivers\WMILIB.SYS
0x8d8a476a82d0      msisadvr.sys      0xf80155fb0000      0xb000 C:\Windows\System32\drivers\msisadvr.sys
0x8d8a476a83f0      pci.sys      0xf8015b750000      0x8c000 C:\Windows\System32\drivers\pci.sys
0x8d8a476a85c0      tpm.sys      0xf8015b7e0000      0x57000 C:\Windows\System32\drivers\tpm.sys
0x8d8a476a8950      aswELam.sys      0xf80155fd0000      0xa000 C:\Windows\system32\drivers\aswELam.sys
0x8d8a476a8b10      intelpep.sys      0xf8015b840000      0x86000 C:\Windows\System32\drivers\intelpep.sys
0x8d8a476a8cf0      WindowsTrustedRT.sys      0xf80155fe0000      0x18000 C:\Windows\system32\drivers\WindowsTrustedRT.sys
0x8d8a476a9010      IntelPMT.sys      0xf8015b8d0000      0x13000 C:\Windows\System32\drivers\IntelPMT.sys
0x8d8a476a91e0      WindowsTrustedRTProx
y.sys      0xf8015b8f0000      0xb000 C:\Windows\System32\drivers\WindowsTrustedRTProxy.sys
0x8d8a476a93d0      pcw.sys      0xf8015b900000      0x16000 C:\Windows\System32\drivers\pcw.sys
0x8d8a476b9c90      vdrvroot.sys      0xf8015b920000      0x1c000 C:\Windows\System32\drivers\vdrvroot.sys
0x8d8a476b9590      pdc.sys      0xf8015b940000      0x31000 C:\Windows\system32\drivers\pdc.sys
```

Analysis:

Suspicious modules are located under -

Suspicious paths (e.g., outside standard C:\Windows\System32\ or DriverStore\)

Unusual names

PRM.sys

- Path:
C:\Windows\System32\DriverStore\FileRepository\prm.inf_amd64...
4...
- The path is slightly unusual, but it can still be legitimate—DriverStore is used for driver packages.
- Recommend verifying the digital signature and hash of this file.

WppRecorder.sys

- This can be part of Windows software trace preprocessor (WPP). Usually legit, but confirm it's not tampered.

IntelPPM.sys, IntelLPT.sys

- Look okay, but always good to verify if they come from Intel and are digitally signed.

6. Are there any processes running with unusual parent-child relationships?

Command:

rekall -f memory_memdumb.mem pstree

```
\Program Files\Rekall> rekall -f "C:\Users\lenovo\Downloads\Programs\memdump.mem" pstree
25-04-04 19:54:32,206:WARNING:rekall.1:Inventory for repository "http://profiles.rekall-forensic.com" seems malformed. Are you behind a captive portal or
oxy? If this is a custom repository, did you forget to create an inventory? You must use the tools/profiles/build_profile_repo.py tool with the --inventor
flag.
```

```
25-04-04 19:54:32,206:WARNING:rekall.1:Repository http://profiles.rekall-forensic.com will be disabled.
```

PROCESS	ppid	thd_count	hnd_count	create_time
0x8d8a476ff040 System (4)	0	445	-	2025-03-25 19:17:15Z
0x8d8a478d6040 Secure System (204)	4	0	-	2025-03-25 19:17:08Z
0x8d8a478e8040 Registry (236)	4	4	-	2025-03-25 19:17:08Z
0x8d8a54277040 smss.exe (880)	4	2	-	2025-03-25 19:17:15Z
0x8d8a57f13080 MemCompression (3604)	4	66	-	2025-03-25 19:17:27Z
0x8d8a61d7e100 csrss.exe (1240)	1000	17	-	2025-03-25 19:17:24Z
0x8d8a647ee100 wininit.exe (1356)	1000	2	-	2025-03-25 19:17:26Z
0x8d8a577790c0 services.exe (1428)	1356	9	-	2025-03-25 19:17:26Z
0x8d8a5e78a0c0 AISControlServ (900)	1428	2	-	2025-03-25 19:17:29Z
0x8d8a5e79b0c0 mDNSResponder. (1348)	1428	2	-	2025-03-25 19:17:29Z
0x8d8a64922140 csrss.exe (1376)	1348	16	-	2025-03-25 19:17:26Z
0x8d8a649be100 winlogon.exe (1500)	1348	5	-	2025-03-25 19:17:26Z
0x8d8a578240c0 fontdrvhost.ex (1696)	1500	5	-	2025-03-25 19:17:26Z
0x8d8a579ce0c0 dwm.exe (2432)	1500	34	-	2025-03-25 19:17:26Z
0x8d8a5daac0c0 userinit.exe (6632)	1500	0	-	2025-03-25 19:17:28Z
0x8d8a5dbbd0c0 explorer.exe (6708)	6632	196	-	2025-03-25 19:17:28Z
0x8d8a6d942080 OneDrive.exe (3248)	6708	46	-	2025-03-25 19:17:42Z
0x8d8a6e5f6080 IDMan.exe (12296)	6708	5	-	2025-03-25 19:17:47Z
0x8d8a6d9980c0 vgtray.exe (20916)	6708	4	-	2025-03-25 19:17:42Z
0x8d8a6bc66080 browser_assist (21144)	6708	8	-	2025-03-25 19:17:44Z
0x8d8a6b8e4080 browser_assist (9004)	21144	7	-	2025-03-25 19:17:44Z
0x8d8a6d9020c0 SecurityHealth (21148)	6708	3	-	2025-03-25 19:17:40Z
0x8d8a6bf8e0c0 uTorrent.exe (21192)	6708	40	-	2025-03-25 19:17:45Z
0x8d8a6b2f3080 utorrentie.exe (21556)	21192	7	-	2025-03-25 19:17:45Z
0x8d8a6b6b9080 utorrentie.exe (21612)	21192	7	-	2025-03-25 19:17:46Z
0x8d8a6dc220c0 msedgewebview2 (22216)	21612	54	-	2025-03-25 19:17:46Z
0x8d8a6dc56080 msedgewebview2 (22544)	22216	7	-	2025-03-25 19:17:46Z
0x8d8a6e0df080 msedgewebview2 (22832)	22216	51	-	2025-03-25 19:17:46Z
0x8d8a6e0e0080 msedgewebview2 (22840)	22216	18	-	2025-03-25 19:17:46Z
0x8d8a6e0da080 msedgewebview2 (22880)	22216	7	-	2025-03-25 19:17:46Z
0x8d8a6e291080 msedgewebview2 (23012)	22216	14	-	2025-03-25 19:17:46Z
0x8d8a6e293100 msedgewebview2 (23072)	22216	23	-	2025-03-25 19:17:46Z
0x8d8a6f4b61c0 msedgewebview2 (26280)	22216	15	-	2025-03-25 19:18:18Z
0x8d8a5b2dd080 RtkAudUService (21344)	6708	9	-	2025-03-25 19:17:41Z
0x8d8a6b6b8080 msedge.exe (21588)	6708	52	-	2025-03-25 19:17:46Z

Analysis:

Suspicious Findings

1. uTorrent.exe and utorrentie.exe

- Location in tree:
 - uTorrent.exe (PID: 21162) appears to spawn:
 - utorrentie.exe (PID: 21163)
 - Concern:
 - While uTorrent is a known torrent client, it's often abused for delivering malware.
 - If this was unexpected behavior or if this system shouldn't be running torrent software, this is a red flag.
 - Check whether these binaries are digitally signed and located in their expected directories.
-

2. Multiple msedgewebview2.exe Instances

- Child of: uTorrent.exe
- Concern:
 - It's odd that a torrent client is spawning many instances (PID 22280, 22284, etc.) of msedgewebview2.exe, a process related to Microsoft Edge's WebView2 runtime.
 - WebView2 is often used by legitimate apps, but this pattern could indicate misuse or even a loader for malicious scripts via a browser component.

- o Analyze these binaries further for suspicious network or file activity.
-

3. browser_assist.exe

- Parent: browser_assist is spawned under a browser tree, but:
 - o The name sounds generic and could be malware posing as an assistant tool.
 - o Check file path and digital signature — malicious browser helpers often use similar names.

7. Were there any unusual or injected code regions in memory?

```

C:\Program Files\Rekal> rekall -f "C:\Users\lenovo\Downloads\Programs\memdump.mem" pstree
025-04-04 19:54:32,206:WARNING:rekall.1:Inventory for repository "http://profiles.rekall-forensic.com" seems malformed. Are you behind a captive portal or
proxy? If this is a custom repository, did you forget to create an inventory? You must use the tools/profiles/build_profile_repo.py tool with the --inventor
y flag.
025-04-04 19:54:32,206:WARNING:rekall.1:Repository http://profiles.rekall-forensic.com will be disabled.

C:\Program Files\Rekal> rekall -f "C:\Users\lenovo\Downloads\Programs\memdump.mem" vadinfo
usage: rekall [-p PROFILE] [-v] [-q] [--debug] [--output_style {concise,full}]
              [--logging_level {DEBUG,INFO,WARNING,ERROR,CRITICAL}]
              [--log_domain [{PageTranslation} [{PageTranslation} ...]]]
              [--pager PAGER] [--paging_limit PAGING_LIMIT]
              [--colors {auto,yes,no}] [-F FORMAT]
              [--plugin [PLUGIN [PLUGIN ...]]] [-h] [--cache CACHE]
              [--repository_path [REPOSITORY_PATH [REPOSITORY_PATH ...]]]
              [-f FILENAME] [--buffer_size BUFFER_SIZE] [--output OUTPUT]
              [--max_collector_cost MAX_COLLECTOR_COST] [--home HOME]
              [--logging_format LOGGING_FORMAT]
              [--performance {normal,fast,thorough}] [--live LIVE]
              [--cpu_quota CPU_QUOTA] [--load_quota LOAD_QUOTA] [--dtb DTB]
              [-o FILE_OFFSET] [--ept EPT [EPT ...]] [--timezone TIMEZONE]
              [--cache_dir CACHE_DIR]
              [--highlighting_style {default,emacs,friendly,colorful,autumn,murphy,manni,monokai,perldoc,pastie,borland,trac,native,fruity,bw,vim,vs,tango,rr
t,xcode,igor,paraiso-light,paraiso-dark,lovelace,algol,algol_nu,arduino,rainbow_dash,abap}]
              [--name_resolution_strategies [{Module,Symbol,Export} [{Module,Symbol,Export} ...]]]
              [--autodetect_build_local_tracked [AUTODETECT_BUILD_LOCAL_TRACKED [AUTODETECT_BUILD_LOCAL_TRACKED ...]]]
              [--pagefile [PAGEFILE [PAGEFILE ...]]]
              [--autodetect {nt_index,pe,rsds,windows_kernel_file,linux_index,linux,osx,ntfs,tsk} [{nt_index,pe,rsds,windows_kernel_file,linux_index,linux,os
x,ntfs,tsk} ...]]
              [--autodetect_threshold AUTODETECT_THRESHOLD]
              [--autodetect_build_local {full,basic,none}]
              [--autodetect_scan_length AUTODETECT_SCAN_LENGTH] [--version]
              [--agent_configuration AGENT_CONFIGURATION] [-]
rekall: error: unrecognized arguments: vadinfo

```

Command:

rekall -f memory_memdump.mem vadinfo

Analysis:

Manually mapped code sections in memory often indicate malware execution, as attackers use this technique to run malicious code without leaving traces on disk.

8. Are there any unlinked or hidden executable files in memory?

Command:

`rekall -f memory_memdumb.mem orphan_threads`

```
C:\Users\lenovo>cd C:\Program Files\Rekall
C:\Program Files\Rekall> rekall -f "C:\Users\lenovo\Downloads\Programs\memdump.mem" malfind
2025-04-10 23:35:11.529:WARNING:rekall.1:Inventory for repository "http://profiles.rekall-forensic.com" seems malformed. Are you behind a captive portal or proxy? If this is a custom repository, did you forget to create an inventory? You must use the tools/profiles/build_profile_repo.py tool with the --inventory flag.
2025-04-10 23:35:11.530:WARNING:rekall.1:Repository http://profiles.rekall-forensic.com will be disabled.
*****1740
Process: ms-teams.exe Pid: 1740 Address: 0x7ff918160000
Vad Tag: VadS Protection: EXECUTE_READWRITE
PrivateMemory: 1, Protection: 6

0x7ff918160000 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ..... vad_0x7ff918160000
0x7ff918160010 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0x7ff918160020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0x7ff918160030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....

----- vad_0x7ff918160000 -----
0x7ff918160000 0x0 0000          add byte ptr [rax], al
0x7ff918160002 0x2 0000          add byte ptr [rax], al
0x7ff918160004 0x4 0000          add byte ptr [rax], al
0x7ff918160006 0x6 0000          add byte ptr [rax], al
0x7ff918160008 0x8 0000          add byte ptr [rax], al
0x7ff91816000a 0xa 0000          add byte ptr [rax], al
0x7ff91816000c 0xc 0000          add byte ptr [rax], al
0x7ff91816000e 0xe 0000          add byte ptr [rax], al
0x7ff918160010 0x10 0000         add byte ptr [rax], al
0x7ff918160012 0x12 0000         add byte ptr [rax], al
0x7ff918160014 0x14 0000         add byte ptr [rax], al
0x7ff918160016 0x16 0000         add byte ptr [rax], al
0x7ff918160018 0x18 0000         add byte ptr [rax], al
0x7ff91816001a 0x1a 0000         add byte ptr [rax], al
0x7ff91816001c 0x1c 0000         add byte ptr [rax], al
0x7ff91816001e 0x1e 0000         add byte ptr [rax], al
0x7ff918160020 0x20 0000         add byte ptr [rax], al
0x7ff918160022 0x22 0000         add byte ptr [rax], al
0x7ff918160024 0x24 0000         add byte ptr [rax], al
0x7ff918160026 0x26 0000         add byte ptr [rax], al
0x7ff918160028 0x28 0000         add byte ptr [rax], al
```

Analysis:

1. EXECUTE_READWRITE permissions (RWX) are uncommon for legitimate code and highly indicative of:
 - o Code injection
 - o Reflective DLL loading
 - o Shellcode
2. Private memory: It's not mapped to a file, suggesting the executable content is:
 - o Not loaded from disk

- o Potentially injected directly into memory
3. ms-teams.exe is a known legit process, making it a good candidate for *process hollowing* or *code injection* by malware trying to hide.

Yes, this memory region is very likely:

- Unlinked from any legitimate file on disk
- Suspicious
- Possibly injected

9. Are there any suspicious kernel hooks or modifications?

Command:

rekall -f memory_memdumb.mem ssdt

Analysis:

```
C:\Program Files\Rekall> rekall -f "C:\Users\lenovo\Downloads\Programs\memdump.mem" orphan ssdt
2025-04-04 19:57:54,354:WARNING:rekall.1:Inventory for repository "http://profiles.rekall-forensic.com" seems malformed. Are you behind a captive portal or
proxy? If this is a custom repository, did you forget to create an inventory? You must use the tools/profiles/build_profile_repo.py tool with the --inventor
y flag.
2025-04-04 19:57:54,354:WARNING:rekall.1:Repository http://profiles.rekall-forensic.com will be disabled.
usage: rekall [-p PROFILE] [-v] [-q] [--debug] [--output_style {concise,full}]
              [--logging_level {DEBUG,INFO,WARNING,ERROR,CRITICAL}]
              [--log_domain [{PageTranslation} [{PageTranslation} ...]]]
              [--pager PAGER] [--paging_limit PAGING_LIMIT]
              [--colors {auto,yes,no}] [-F FORMAT]
              [--plugin [PLUGIN [PLUGIN ...]]] [-h] [--cache CACHE]
              [--repository_path [REPOSITORY_PATH [REPOSITORY_PATH ...]]]
              [-f FILENAME] [--buffer_size BUFFER_SIZE] [--output OUTPUT]
              [--max_collector_cost MAX_COLLECTOR_COST] [--home HOME]
              [--logging_format LOGGING_FORMAT]
              [--performance {normal,fast,thorough}] [--live LIVE]
              [--cpu_quota CPU_QUOTA] [--load_quota LOAD_QUOTA]
              [-o FILE_OFFSET] [--ept EPT [EPT ...]] [--timezone TIMEZONE]
              [--cache_dir CACHE_DIR]
              [--highlighting_style {default,emacs,friendly,colorful,autumn,murphy,manni,monokai,perldoc,pastie,borland,trac,native,fruity,bw,vim,vs,tango,r
t,xcode,igor,paraiso-light,paraiso-dark,lovelace,algol,algol_nu,arduino,rainbow_dash,abap}]
              [--name_resolution_strategies [{Module,Symbol,Export} [{Module,Symbol,Export} ...]]]
              [--autodetect_build_local_tracked [AUTODETECT_BUILD_LOCAL_TRACKED [AUTODETECT_BUILD_LOCAL_TRACKED ...]]]
              [--pagefile [PAGEFILE [PAGEFILE ...]]]
              [--autodetect {nt_index,pe,rsds,windows_kernel_file,linux_index,linux,osx,ntfs,tsk} [{nt_index,pe,rsds,windows_kernel_file,linux_index,linux,os
x,ntfs,tsk} ...]]
              [--autodetect_threshold AUTODETECT_THRESHOLD]
              [--autodetect_build_local {full,basic,none}]
              [--autodetect_scan_length AUTODETECT_SCAN_LENGTH] [--version]
              [--agent_configuration AGENT_CONFIGURATION] [-]
              [--verbosity VERBOSITY]
rekall: error: unrecognized arguments: orphan
```

System service call hooks suggest possible rootkit activity, as attackers use them to intercept, modify, or hide malicious operations from security tools.

10. Are there any suspicious handles open by running processes?

Command:

```
C:\Program Files\Rekal> rekall -f "C:\Users\lenovo\Downloads\Programs\memdump.mem" handles
2025-04-04 20:02:40,768:WARNING:rekall.1:Inventory for repository "http://profiles.rekall-forensic.com" seems malformed. Are you behind a captive portal or proxy? If this is a custom repository, did you forget to create an inventory? You must use the tools/profiles/build_profile_repo.py tool with the --inventory flag.
2025-04-04 20:02:40,768:WARNING:rekall.1:Repository http://profiles.rekall-forensic.com will be disabled.
_OBJECT_HEADER _EPROCESS      name      pid      handle      access      obj_type      details
-----
C:\Program Files\Rekal> rekall -f "C:\Users\lenovo\Downloads\Programs\memdump.mem" hashdump
usage: rekall [-p PROFILE] [-v] [-q] [--debug] [--output_style {concise,full}]
              [--logging_level {DEBUG,INFO,WARNING,ERROR,CRITICAL}]
              [--log_domain [{PageTranslation} [{PageTranslation} ...]]]
              [--pager PAGER] [--paging_limit PAGING_LIMIT]
              [--colors {auto,yes,no}] [-F FORMAT]
              [--plugin [PLUGIN [PLUGIN ...]]] [-h] [--cache CACHE]
              [--repository_path [REPOSITORY_PATH [REPOSITORY_PATH ...]]]
              [-f FILENAME] [--buffer_size BUFFER_SIZE] [--output OUTPUT]
              [--max_collector_cost MAX_COLLECTOR_COST] [--home HOME]
              [--logging_format LOGGING_FORMAT]
              [--performance {normal,fast,thorough}] [--live LIVE]
              [--cpu_quota CPU_QUOTA] [--load_quota LOAD_QUOTA] [--dtb DTB]
              [-o FILE_OFFSET] [--ept EPT [EPT ...]] [--timezone TIMEZONE]
              [--cache_dir CACHE_DIR]
              [--highlighting_style {default,emacs,friendly,colorful,autumn,murphy,manni,monokai,perldoc,pastie,borland,trac,native,fruity,bw,vim,vs,tango,rrt,xcode,igor,paraiso-light,paraiso-dark,lovelace,algol,algol_nu,arduino,rainbow_dash,abap}]
              [--name_resolution_strategies [{Module,Symbol,Export} [{Module,Symbol,Export} ...]]]
              [--autodetect_build_local_tracked [AUTODETECT_BUILD_LOCAL_TRACKED [AUTODETECT_BUILD_LOCAL_TRACKED ...]]]
              [--pagefile [PAGEFILE [PAGEFILE ...]]]
              [--autodetect {nt_index,pe,rsds,windows_kernel_file,linux_index,linux,osx,ntfs,tsk} [{nt_index,pe,rsds,windows_kernel_file,linux_index,linux,osx,ntfs,tsk} ...]]
              [--autodetect_threshold AUTODETECT_THRESHOLD]
              [--autodetect_build_local {full,basic,none}]
              [--autodetect_scan_length AUTODETECT_SCAN_LENGTH] [--version]
              [--agent_configuration AGENT_CONFIGURATION] [-]
rekall: error: unrecognized arguments: hashdump
```

rekall -f memory_memdump.mem handles

Analysis:

Suspicious open handles may indicate malware interacting with system resources, such as accessing sensitive files, injecting code into processes, or manipulating the registry.

Submitted to:

Prof: Rizwan Ur Rehman

Submitted by:

Group 6:

Atharv Tyagi (24BCY10336)

Garima Singh (24BCY10342)

Ashqua Islam (24BCY10345)

Krishna Shivhare (24BCY10360)